ACADEMIC PROJECT DOCUMENTATION

Student Enrollment & Course Management System

Phase 5: Deployment, Documentation & Final Presentation

Platform: Salesforce Developer Edition (Lightning Experience)

PROJECT TEAM

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Document scope: Phase 5 only. Version 1.0, prepared September 2026.

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Note. Section 1 is the cover page. **Status key. Completed / evidenced** = shown in the supplied project screenshots. **Documented** = described in the project guide without a screenshot. **Pending** = not yet performed or not evidenced. **Proposed** = suggested, not implemented. **Requires confirmation** = missing or inconsistent information. No deployment, test result or team contribution is recorded here unless the sources support it.

2. Project Overview

The Student Enrollment & Course Management System (SECMS) is a Salesforce application that replaces manual, spreadsheet-based enrollment handling with a single system. It lets administrators, enrollment officers and instructors manage student records, courses, enrollments and instructor assignments, with validation, approvals and reporting. The problems addressed are scattered enrollment data and delayed reporting.

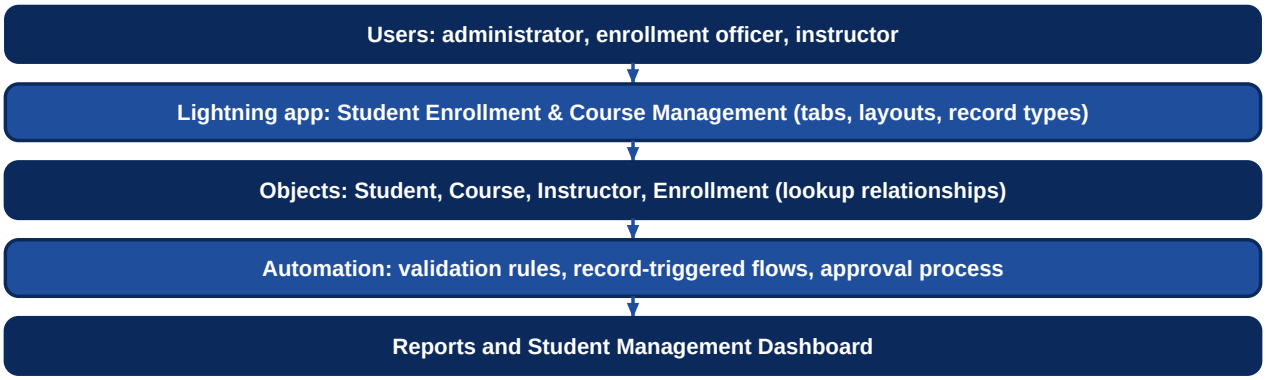


Diagram 1. *Layers of the SECMS solution as described in the project file.*

Main Features

Feature	Description	Status
Custom objects	Student, Course, Instructor and Enrollment, linked by lookups from Enrollment	Configured (project file)
Lightning app and tabs	App "Student Enrollment & Course Management" with tabs for the four objects	Evidenced
Record types and layouts	New Enrollment and Re-Enrollment record types with separate layouts	Evidenced
Validation rules	Student email domain; fees paid before approval	Evidenced (rule screens)
Automation	Flows for enrollment date, approval actions, instructor assignment and follow-up task	Evidenced (flow screens); results untested
Approval process	Enrollment requests routed to a Training Manager step	Evidenced (setup screens)
Reports and dashboard	Four reports and the Student Management Dashboard	Evidenced
Security	Profiles, permission sets and sharing defaults	Partly documented; see Section 5

3. Deployment Preparation

The table separates preparation tasks that the evidence shows as complete from those that remain planned.

Preparation task	Status	Basis / remaining action
Review objects and fields	Configured; final review pending	Field lists in the project file. Confirm API names (for example the Student status field) before packaging
Verify object relationships	Configured; verification pending	Three required lookups from Enrollment shown in setup screens. Confirm by opening related records

Preparation task	Status	Basis / remaining action
Review validation rules	Configured; review pending	Two rules shown. Review the email rule (blank and partial matches)
Review record-triggered flows	Configured; review pending	Flow screens shown. Issue I-01 (update step in the approval flow) is unresolved
Check page layouts	Partly evidenced	Enrollment layouts shown. Student, Course and Instructor layouts not shown
Check Lightning record pages	Pending	No Lightning App Builder evidence supplied
Verify permissions and security	Partly evidenced	App assigned to System Administrator and Standard User. Other settings documented only
Complete functional and integration testing	Pending	Test cases were prepared, but no executed results were supplied
Prepare deployment documentation	Completed in this document	Section 4

4. Salesforce Deployment

Status: deployment to another org or to production is not confirmed. The project was built in a Salesforce Developer Edition org. No deployment package, change set or CLI output was supplied.

4.1 Deployment Approaches

Approach	How it works	Suitability for SECMS
Change sets	Point-and-click movement of metadata through outbound and inbound change sets over a deployment connection	Works between related orgs such as a sandbox and its production org. One source states that metadata cannot be moved by change sets from a Developer Edition org to a sandbox or production org. Not suitable unless the team has a sandbox and production pair
Salesforce CLI and Metadata API	Retrieve metadata from a source org into a project using a manifest (package.xml), then deploy it to a target org	Suitable if the team wants a copy in another org, including another Developer Edition org. Requires CLI setup and login to both orgs
Sandbox to production	Build and test in a sandbox, then promote to production	Not applicable to the current environment. Appropriate for a real institution
Use the Developer Edition org as the final environment	The configured org is presented and demonstrated directly	Matches the actual environment. This is the recommended approach for the college submission

4.2 General Deployment Process

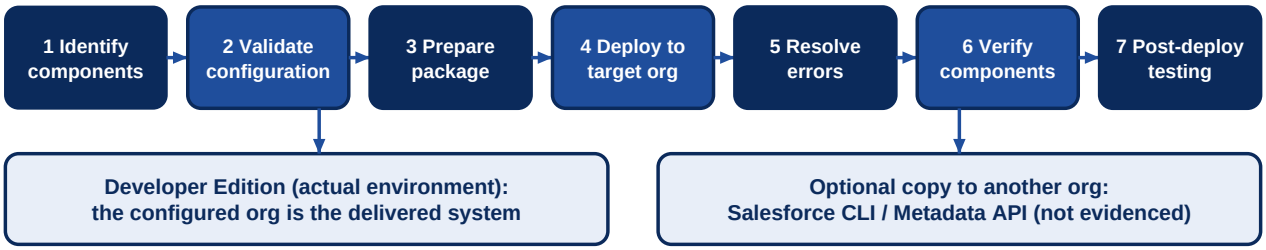


Diagram 2. *General deployment process, and the two routes relevant to a Developer Edition project.*

Step	Action	SECMS notes
1	Identify the required components	Use the component inventory in Section 4.3
2	Validate the configuration	Check that dependent items exist (fields before layouts, email template before email alert). Use a validate or dry-run option where available
3	Prepare the deployment package	Change set components, or a package.xml manifest for CLI deployment
4	Deploy to the target org	Only after the target org and its access are confirmed
5	Resolve deployment errors	Read the error list; typical causes are missing dependencies, inactive flows referencing missing items and missing users
6	Verify deployed components	Open each object, layout, flow and report in the target org
7	Perform post-deployment testing	Repeat the checks in Section 5 and the test cases in Section 8

Indicative command sequence (Salesforce CLI, not executed for this project):

```
sf org login web --alias source-org
sf project retrieve start --manifest manifest/package.xml --target-org source-org
sf org login web --alias target-org
sf project deploy start --manifest manifest/package.xml --target-org target-org --dry-run
```

Command names should be checked against the current Salesforce CLI documentation before use.

4.3 Component Inventory

Component group	Items (from the project file)	Status
Objects and fields	Student, Course, Instructor, Enrollment and their fields	Configured
Relationships	Enrollment lookups to Student, Course and Instructor	Configured
Record types and layouts	New Enrollment, Re-Enrollment and matching layouts; other object layouts	Enrollment evidenced
Tabs and app	Tabs for the four objects; Lightning app	Evidenced
Automation	Two validation rules; flows; approval process, field updates, email alert and template	Configured
Reports and dashboard	Students by Status, Enrollments by Course, Pending Enrollment Approvals, Revenue Report - Fees Paid; Student Management Dashboard	Evidenced
Security	Profiles, permission sets, sharing settings	Documented / partly evidenced

Environment dependencies to check. Metadata deployment does not copy data. Sample records must be loaded again in the target org. Some items depend on specific users: the approval process uses a named Training Manager user, the follow-up task flow looks up an admin user by username, and the rejection email alert has selected recipients. These users must exist in the target org or the items must be edited.

5. Final System Verification

The checklist marks each item according to the evidence available. **Observed** means a screenshot shows the configuration or data, but no formal test outcome was recorded. **Pending** means it still needs to be performed. No item is marked as passed.

Function	What to verify	Evidence available	Mark
Student management	Create, view and update student records	Students tab and layout exist; a report lists 4 student records	Observed; test pending
Course management	Create and view courses; category and fee	A report lists courses (Computer Science, Electronic Communications & Engineering)	Observed; test pending
Enrollment processing	Create enrollments with the correct record type and layout	Record types and layouts shown; 2 enrollment records in a report	Observed; test pending
Instructor assignment	Category decides the instructor (Technical A, Language B, Non-Technical C)	Active flow canvas shown; no assigned instructor value shown	Configuration observed; result pending
Enrollment status management	Status moves through Requested, Approved, Rejected	Approval process and field updates shown; both records show Requested	Configuration observed; result pending
Validation rules	Invalid email and unpaid approval are blocked	Rule setup screens shown; no triggered error screen supplied	Configuration observed; test pending
Record-triggered flows	Flows fire and update the right fields	Flow screens shown. Enrollment Date is filled on both enrollments in a report, which is consistent with the date flow but not proof	Partly observed; I-01 unresolved
Reports and dashboards	Reports run and the dashboard shows components	Report and dashboard screenshots with data	Observed; data accuracy pending (I-03)
User access and security	Each role sees only permitted data	App assigned to two profiles; other settings not shown	Pending

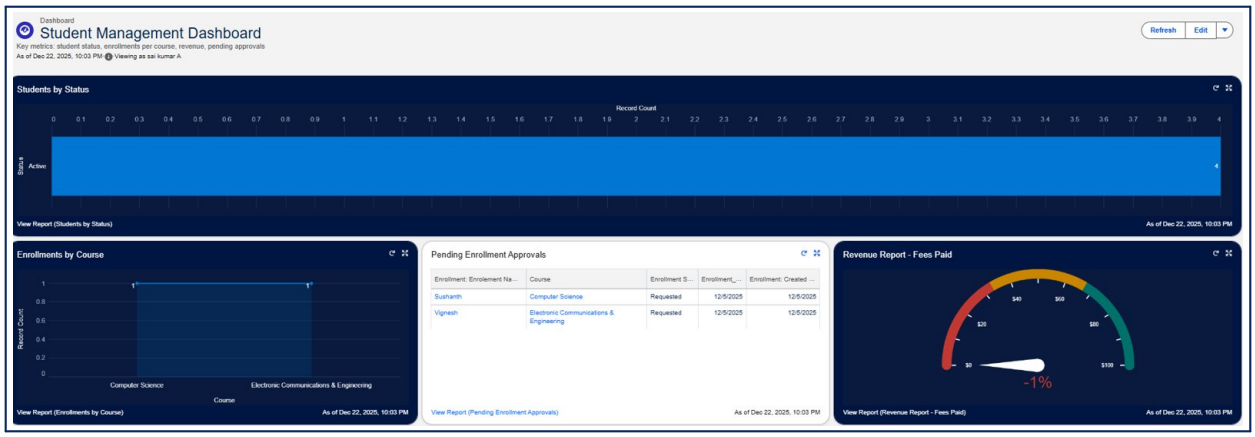


Figure 1. *Student Management Dashboard: Students by Status, Enrollments by Course, Pending Enrollment Approvals and Revenue Report - Fees Paid components.*

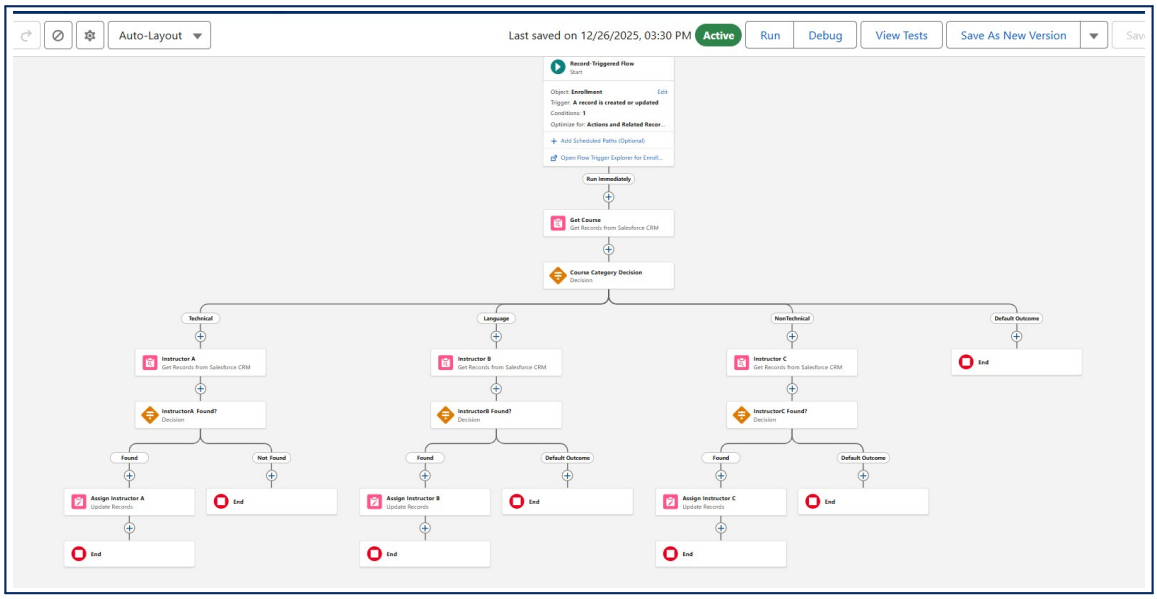


Figure 2. *AutoAssign_Instructor flow (Active): branches for Technical, Language and NonTechnical courses.*

6. Complete Project Documentation

The final submission should include the documents below. The table shows where each is available and what is still needed. Details of earlier phases are not repeated here.

Document	Source in project file	Status
Project introduction	Sections 1.1 and 1.2	Available
Problem statement	Section 2.1 (PS-1 enrollment officer, PS-2 administrator)	Available
Objectives	Purpose list in Section 1.2	Available
Functional requirements	FR-1 to FR-5 and NFR-1 to NFR-6	Available
Object and field definitions	Milestones 2 and 3 tables	Available; confirm API names
Relationship diagram	Diagram 2 below and architecture image	Available
Business rules	Validation rules, approval process, category-to-instructor rules	Available
Process flow diagrams	Data flow diagram, flow canvas	Available
UI screenshots	Tabs, app, layouts, record types	Partly available
Backend configuration	Milestones on flows, approvals, reports, dashboards	Available
Testing results	No executed results supplied	Pending
Security configuration	Milestone on profiles, permission sets, sharing	Documented; screenshots pending
Deployment steps	Section 4 of this document	Prepared; not executed
Conclusion and future enhancements	Project file Sections 9 and 10; Section 10 here	Available

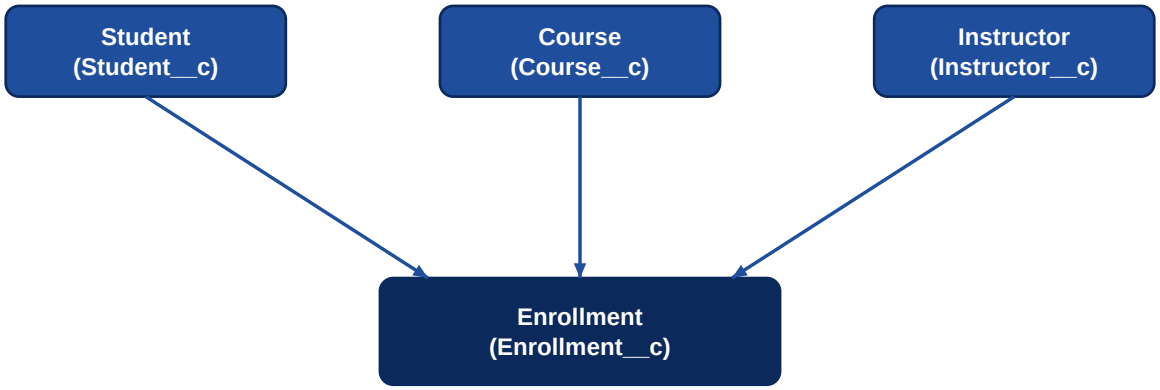


Diagram 3. *Object relationship diagram. Each Enrollment stores a Student, a Course and an Instructor through lookup fields.*

7. User Guide

This guide is for an authorised user. Steps that rely on standard Salesforce behaviour without a supplied screenshot are marked *(standard)*. Only navigation items visible in the screenshots are named.

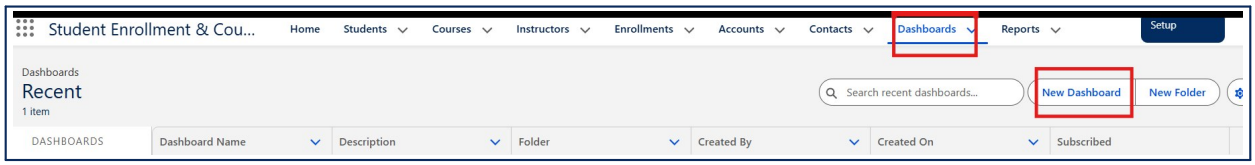


Figure 3. *The app navigation bar: Home, Students, Courses, Instructors, Enrollments, Accounts, Contacts, Dashboards and Reports, with the App Launcher icon at the far left.*

Step-by-step

Step	Instruction
1. Log in to Salesforce	Open the login page of your Salesforce Developer Edition org and enter your username and password. *(standard)*
2. Navigate to the application	Click the App Launcher icon (nine dots) at the top left and choose **Student Enrollment & Course Management**. The navigation bar above appears.
3. Create or view student records	Click **Students**. Open a record from the list to view it. To add one, use the New button on the list page, complete the fields and click Save. *(standard)* The email must use the institutional domain (Figure: validation rule).
4. Create or view course records	Click **Courses**. View records from the list, or use New to add a course with name, category, duration and fee. *(standard)*
5. Create enrollment records	Click **Enrollments** and choose New. If asked, select the record type **New Enrollment** (first-time) or **Re-Enrollment**. Select the Student and Course, set the Enrollment Status and Save. Instructor is a required field on the layout.
6. View instructor assignments	Open an enrollment and read the **Instructor** field. The flow may assign it from the course category. You can also open **Instructors** and look at the Enrollments related list if it is on the layout.
7. Check enrollment status	Open the enrollment and read **Enrollment Status** (for example Requested, Approved or Rejected). The Pending Enrollment Approvals report lists enrollments with status Requested.
8. Access reports and dashboards	Click **Reports** or **Dashboards** in the navigation bar, then open the Student Management Dashboard or one of the four reports.

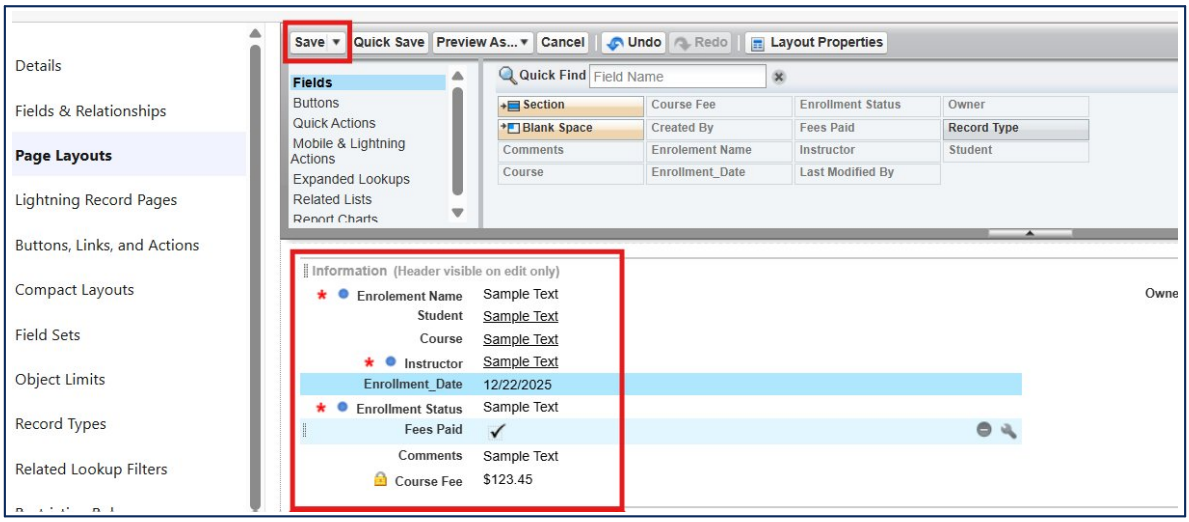


Figure 4. *What the enrollment form looks like in the New Enrollment layout editor: Enrolement Name, Student, Course, Instructor, Enrollment_Date, Enrollment Status, Fees Paid, Comments and Course Fee. Red asterisks mark required fields.*

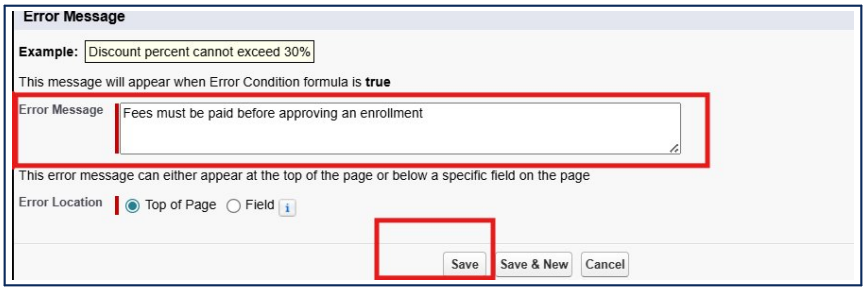


Figure 5. *Message users see when approving an enrollment without paid fees: "Fees must be paid before approving an enrollment".*

8. Final Testing and Results

Actual results come only from the supplied evidence. No test execution log was supplied, so **no test is marked Passed**. Status meanings: Observed = configuration or data seen in a screenshot; Pending = not yet tested.

ID	Feature Tested	Expected Result	Actual Result	Status	Remarks
FV-01	Student record creation	Valid student saves	Not recorded	Pending	4 student records exist; creation method unconfirmed
FV-02	Course record creation	Valid course saves	Not recorded	Pending	Courses exist in report
FV-03	Enrollment creation with record type	Correct layout by record type	Layouts configured	Observed	Runtime test pending
FV-04	Instructor assignment by category	A, B or C assigned	Flow active; no result shown	Pending	Mapping per guide and flow
FV-05	Enrollment status update	Status changes; actions run	Two enrollments show Requested	Pending	Approval not evidenced as run
FV-06	Email validation rule	Invalid email blocked	Rule screen only	Pending	No triggered error supplied

ID	Feature Tested	Expected Result	Actual Result	Status	Remarks
FV-07	Fees paid validation rule	Approval blocked when unpaid	Rule screen only	Pending	Interaction with approval to test
FV-08	Enrollment date flow	Date set on creation	Date shown on both records	Observed	Consistent with flow; not confirmed
FV-09	Follow-up task flow	Task created for Requested enrollments	Not recorded	Pending	Task record not shown
FV-10	Reports	Reports show correct counts	Reports display data	Observed	Fee data shows $0.00 (I-03)
FV-11	Dashboard	Components display report data	Dashboard displays four components	Observed	Gauge shows -1% (I-03)
FV-12	Role-based access	Each role limited to permitted data	Not recorded	Pending	Profiles need confirmation
FV-13	Deployment to another org	Components present and working	Not performed	Pending	Not evidenced

9. Known Limitations

The items below are supported by the supplied evidence. Other limitations still need to be confirmed by the team.

Ref	Limitation or unresolved issue	Basis	Status
L-01	No executed test results or test screenshots were supplied, although the project guide asks for them	Project guide Section 7	Open
L-02	The approval flow update step "Set Student Status Active" is shown setting the Comments field to True rather than a student status (I-01)	Flow screenshot	Open, requires confirmation
L-03	Revenue data shows a course fee of $0.00 and the gauge reads -1% (I-03)	Report and dashboard screenshots	Open
L-04	The guide mentions Apex trigger and seat availability checks in testing, but no such configuration appears in the build steps	Project guide Section 7	Requires confirmation
L-05	The guide refers to record type behaviour by course category; the configured record types are New Enrollment and Re-Enrollment	Project guide and screenshots	Requires confirmation
L-06	Enrollment Officer and Instructor profiles are not shown in the app assignment	App assignment screenshot	Requires confirmation
L-07	Lightning record pages, custom list views and duplicate rules are not evidenced	No screenshots	Requires confirmation
L-08	No deployment to another org is evidenced; change sets are not available from a Developer Edition org to a sandbox or production org	Section 4	Environment limitation
L-09	The label "Enrolement Name" is misspelled on the layouts	Layout screenshots	Open

10. Future Enhancements

All items below are proposed future enhancements. None is implemented.

Area	Proposed enhancement	Source
Student self-service	A portal where students view courses and request enrollment	Project file future scope
Payment	Payment gateway integration for fee payment	Project file future scope
Reporting	Analytics-driven insights and additional dashboards, such as enrollment trends	Project file future scope; team proposal
Mobile	Mobile access support	Project file future scope
Automation	Duplicate rules for students and enrollments; seat availability checks; automatic reminders	Team proposal
Notifications	Email or in-app notifications for status changes	Team proposal
User experience	Custom list views, Lightning record pages and role-specific home pages	Team proposal

11. Final Presentation Structure

This is a suggested outline of ten slides. It is a proposal for the team to adapt. Live demonstration steps should be rehearsed in the actual org first.

Slide	Content	Speaker notes
1. Project introduction	Title, team and purpose of SECMS.	Introduce the project and the team. State that it is a Salesforce application for student enrollment and course management.
2. Problem statement	Scattered data, manual processes, delayed reporting.	Explain the two user problems: enrollment officers work with scattered data, and administrators get delayed reports.
3. Objectives	Accurate data, structured workflow, automated approvals, transparent reporting.	List the four goals from the project purpose and connect each to a feature.
4. System architecture	Layers diagram (Diagram 1).	Walk from users to the app, objects, automation and reports. Mention Salesforce Developer Edition.
5. Objects and relationships	Relationship diagram (Diagram 3).	Explain that Enrollment links a student, a course and an instructor.
6. Backend automation	Validation rules, flows, approval process.	Describe each rule and flow briefly, including instructor assignment by category. Say which results are still being verified.
7. UI/UX customization	App navigation, record types, layouts.	Show the navigation bar and the New Enrollment layout.
8. Data migration, testing and security	Migration plan, test cases, access design.	Explain that these were prepared as plans. Report only test results that have been recorded.
9. Deployment and demonstration	Chosen approach; live demo.	State that the Developer Edition org is the delivered environment. Demonstrate creating a student, an enrollment, and viewing the dashboard.

Slide	Content	Speaker notes
10. Results and conclusion	Confirmed work, limitations, future scope.	Summarise what is confirmed, be honest about open items, and present the proposed enhancements.

12. Team Contributions

Name	Role	Verified contribution
AKASH C	Team Lead	To be confirmed by the team
A MOHAMMED IMRAN	Team Member	To be confirmed by the team
SARANRAJ S	Team Member	To be confirmed by the team

Requires confirmation. The sprint backlog in the project file assigns work to "Member1" to "Member4", four placeholders for three named members, and does not say which person each placeholder is. Individual contributions therefore cannot be stated. The team should add verified task assignments before submission.

13. Final Conclusion

SECMS is designed to give a college one Salesforce application for managing students, courses, instructors and enrollments. The project file and screenshots confirm the four custom objects, a Lightning app with tabs, Enrollment record types and layouts, two validation rules, several record-triggered flows, an approval process, four reports and a management dashboard.

Testing evidence is limited: reports and the dashboard display data, but no executed test results, security tests or deployment outputs were supplied. Several items need confirmation, including a flow update step, revenue data, user profiles and the team task assignments. The Developer Edition org is the environment in which the system exists.

Remaining work

- Execute and record the functional, integration and security tests with screenshots.
- Resolve or explain the items in Section 9.
- Confirm user profiles and permissions.
- Add verified team contributions.
- Rehearse the presentation and demonstration in the actual org.

References

- Student Enrollment & Course Management System (SECMS) project documentation and screenshots, supplied working file, Team AKASH C.
- Salesforce. "Deploy Your Community with Change Sets" (Salesforce Developer documentation). https://developer.salesforce.com/docs/atlas.en-us.216.0.communities_dev.meta/communities_dev/networks_migrate_changesets.htm (change sets move metadata between related orgs with a deployment connection).
- Packt Publishing. Salesforce Advanced Administrator Certification Guide, "Deploying changes with change sets" (change set steps and the restriction on Developer Edition orgs). https://m.cover.qq.com/read/1036698362/71

- Gearset. "Salesforce DevOps Center vs change sets." https://gearset.com/blog/devops-center-versus-change-sets/ (change sets only between orgs related to the same production org).
- The sources are older or third-party pages, so deployment options should be checked against current Salesforce Help before use.